

Homework Assignment 2

AWB and Tone Mapping

Digital Image Processing

Deadline: 09:00 AM, 10/27/2025

HW 2(a) Problem Statement

- Auto white balance (AWB) is an important operation for a digital camera. To help you building your own ISP pipeline for your term project, you are asked to implement three AWB methods: 1) Grey world algorithm (25%), 2) White patch algorithm (25%), and 3) An algorithm of your own choice that exploits chromatic adaptation, color constancy, or advanced illumination detection and is more advanced than the first two methods (25%).
- Given five .tif images and corresponding ground-truth illuminants used to capture these images, you are required to compute the angular error between the estimated illuminant (white point) and the ground-truth illuminant in the RGB space and provide an analysis of the results.



References: AWB Methods

Nature of Mapping

Chromatic adaptation map

- J. Von Kries, 1905
- D. Jameson, L.M. Hurvich, 1964
- A. Poirson, B.A. Wandell, 1996
- X. Zhang, B.A. Wandell, 1997
- E.J. Chichilnisky, B.A. Wandell, 1999
- V.C. Smith, J. Pokorny, 1996
- K. Hiraoka, T.W. Parks, 2005

Color constancy *approximation* map

- E.H. Land, 1974
- G. Buchsbaum, 1980
- G. West, M. H. Brill, 1982
- G. Finlayson, M. Drew, B. Funt, 1993
- G. Finlayson, S. Hordley, A. Alsam, 2006
- H. Chong, S. Gortler, T. Zickler, 2007

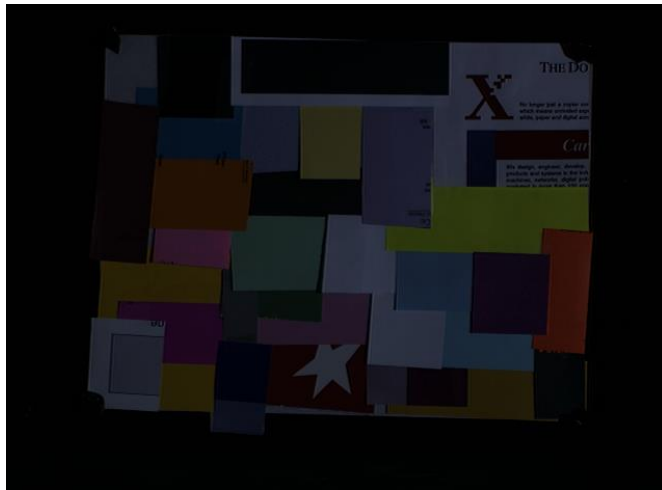
Illuminant Estimation

- E. Land, 1974. [*Grey World*]
- D. Brainard, W. Freeman, 1993. [*Bayesian/Priors*]
- G. Finlayson, S. Hordley, P. Hubel, 2001. [*Color by Correlation*]
- V. Cardei, B. Funt, K. Barnard, 2002. [*Neural Networks*]
- J. van de Weijer, T. Gevers, A. Gijsenij, 2007. [*Grey Edge*]
- A. Chakrabarti, K. Hiraoka, T. Zickler, 2008. [*Spatial Model*]

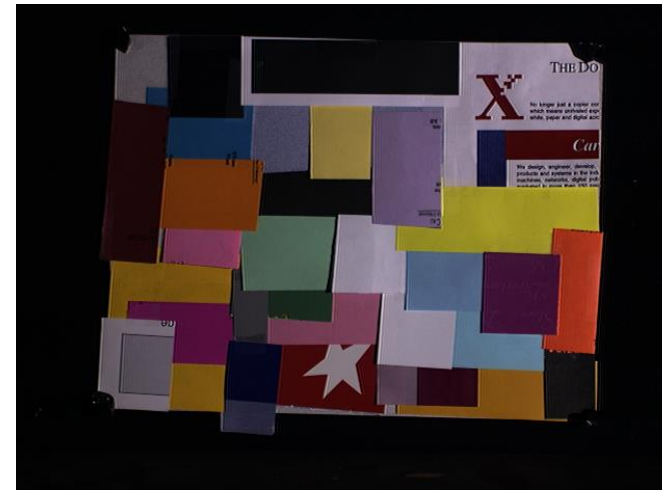
- chromatic adaptation/color constancy
“maps” are parameterized by illuminant
- “two-step” approach: estimate illuminant,
then map colors

HW 2(b) Problem Statement

- Tone mapping, normally a nonlinear operation, is applied to achieve a preferred tone reproduction. In this homework, you are asked to determine a tone mapping curve to transform the white-balanced images you obtained in HW 2(a) to the reference image. Use MSE to evaluate your tone mapping method. (25%)



White-balanced image



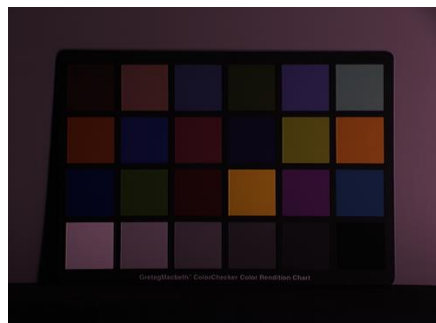
Reference Image

Image Data

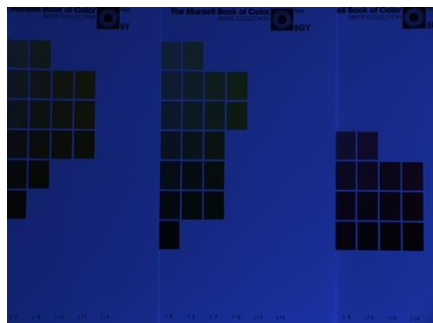
Test
Images



a.tif



b.tif



c.tif



d.tif

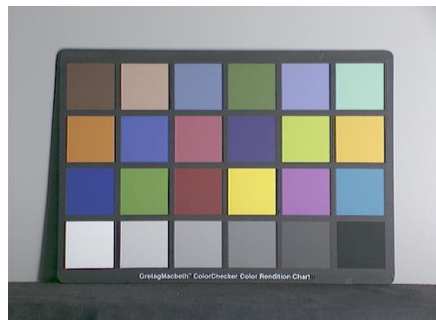


e.tif

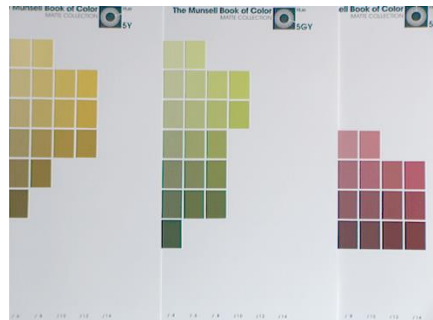
Reference
Images



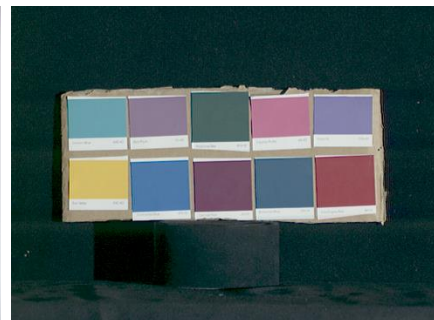
a_reference.tif



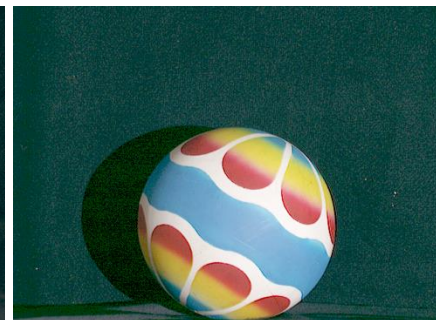
b_reference.tif



c_reference.tif



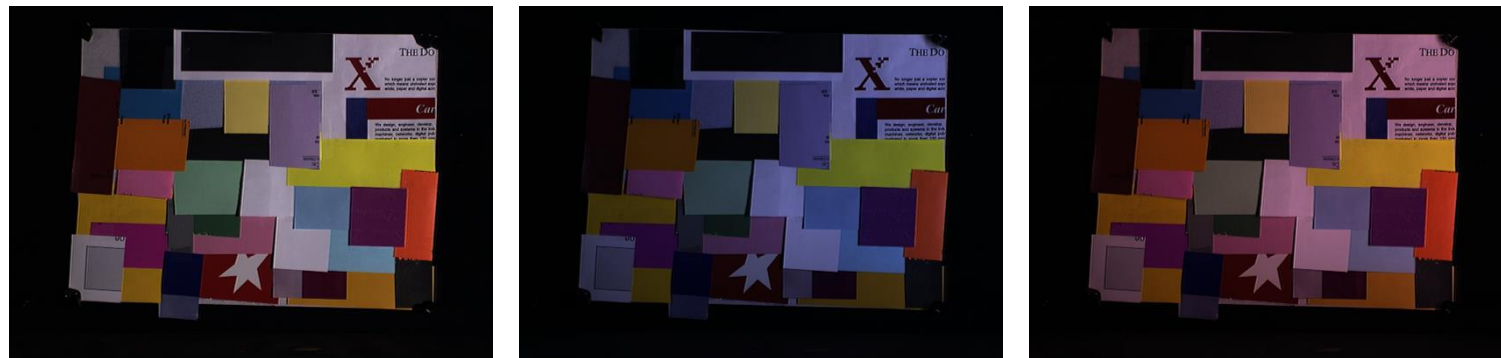
d_reference.tif



e_reference.tif

Dataset Description

- For each scene/illuminant combination in this dataset, there are 2 data files
 1. The image of the scene captured under the illuminant (.tif)
 2. The RGB value of the illuminant (.rgb)
- More information of the data collection procedure is available in: [Test Images for Computational Color Constancy](#). The information is not needed for this homework; however, it may be useful for your term project.
- In essence, AWB algorithms estimate the color of the illuminant and use it to balance the image. Further information of illuminant estimation, which is central to color constancy and AWB can be found in
Finlayson, G. D., & Trezzi, E. (2004, January). *Shades of gray and colour constancy*. In *Color and Imaging Conference* (Vol. 12, pp. 37–41). Society of Imaging Science and Technology.



Images of a scene captured with different illuminants

Grading Policy

- You should provide a detailed description of the techniques employed.
- You should provide a thorough discussion of the results you have obtained.
- You should analyze the strengths and weakness of the employed techniques.
- Your results should be tested against the ground-truths. The score will be based on the quality (correctness) of your results and analysis.
- Your computer code will be tested by TA using a sixth test image.
- Bonus points will be awarded for creative thinking.
- Total points: 100.

Software Package Allowed

- Python 3.8+
- Standard Python library
- Numpy
- Opencv-python

Assignment Requirements

- In part 1) and 2) in HW 2(a), AWB operations should be written by yourself.
- Set your directory structure as follows:
 - HW02_<your ID>/
 - program files (p1.py, p2.ipynb,...)
 - report.pdf
 - images/
 - Images of program output

Assignment Submission Requirements

- Submit to **NTU COOL**
- Do NOT copy homework (code, report, results, etc.) from others